

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output modalities — binary classification labels and abstractive text summaries **[Q82, Q84]** — match the deployment's text-based requirements at the surface level (elicitation rated OF as MODERATE, the only non-HIGH dimension). However, evaluation is via ROUGE n-gram overlap against English reference summaries **[Q98, Q99]** and an English-language human rubric **[Q153]**, neither of which transfers cleanly to Spanish humanitarian reporting. Spanish-tokenization-aware ROUGE or multilingual BERTScore would be needed [evaluation_metric_adaptation], and UN OCHA / Red Cross situation reports follow specific genre conventions (numbered bullets, standardized headers, figures and percentages) that diverge from the abstractive summary style elicited in the benchmark [humanitarian_information_management]. Average timeline-level accuracy **[Q91]** as the extraction metric is generic and portable. The benchmark also does not penalize or evaluate language mismatch between input and output — so a system producing English summaries from Spanish tweets would not be flagged.

ID	Evidence
Q82	"Given a crisis event timeline consisting of n tweets $T = [t_0, t_1, \dots, t_n]$, where t_0 is the seed tweet, i.e. the first observed post that describes the ongoing event, we define two tasks: Timeline Extraction and Timeline Summarization." (p.6)
Q84	"Timeline Summarization: Similar to previous work by Chen et al. (2019), this task aims to generate a summary $\hat{Y} = \{\hat{w}_1, \dots, \hat{w}_k\}$ with sequence of words $\hat{w}_i$ to concisely describe the crisis event and its evolution given the output from the timeline extraction task Textextracted." (p.7)
Q91	"For evaluation, we use the average timeline-level accuracy." (p.7)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)

Strengths

- Both tasks produce text outputs (binary labels, abstractive summaries) **[Q82, Q84]** matching deployment surface modality; multi-reference ROUGE setup **[Q99, Q100]** is methodologically sound and could be replicated against Spanish reference summaries; the four-axis human rubric **[Q106]** is decomposable and adaptable.

ID	Evidence
Q82	"Given a crisis event timeline consisting of n tweets $T = [t_0, t_1, \dots, t_n]$, where t_0 is the seed tweet, i.e. the first observed post that describes the ongoing event, we define two tasks: Timeline Extraction and Timeline Summarization." (p.6)
Q84	"Timeline Summarization: Similar to previous work by Chen et al. (2019), this task aims to generate a summary $\hat{Y} = \{\hat{w}_1, \dots, \hat{w}_k\}$ with sequence of words $\hat{w}_i$ to concisely describe the crisis event and its evolution given the output from the timeline extraction task Textextracted." (p.7)
Q99	"The summarization output of each model is evaluated in a multi-reference setting against the two summaries written by the two workers who agree the most based on the timeline extraction labels." (p.8)
Q100	"We use SacreROUGE (Deutsch and Roth, 2020) to compute multi-reference ROUGE scores." (p.8)

Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
-------------	---

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Surface modality matches: binary classification + abstractive text [Q82, Q84]. However, deployment requires bilingual output (English for OCHA/Red Cross global reporting, Spanish for Ecuadorian government coordination) — the benchmark does not evaluate language-conditioned generation.

OF-2: Not applicable — speech-based output is not required; analyst population is high-literacy and produces written reports [literacy_and_education].

OF-3: The analyst user population has high literacy in Spanish and English [literacy_and_education]; this dimension is not a barrier. Tweet-author population literacy is more variable [WEB-2, WEB-3, WEB-5] but affects input rather than output form.

OF-4: Form mismatches: ROUGE is English-anchored [Q98]; human rubric refers to 'English errors' [Q153]; summary style does not match UN OCHA / Red Cross genre conventions; no evaluation of language-mismatch between Spanish input and required Spanish/English output.

ID	Evidence
Q82	"Given a crisis event timeline consisting of n tweets $T = [t_0, t_1, \dots, t_n]$, where t_0 is the seed tweet, i.e. the first observed post that describes the ongoing event, we define two tasks: Timeline Extraction and Timeline Summarization." (p.6)
Q84	"Timeline Summarization: Similar to previous work by Chen et al. (2019), this task aims to generate a summary $\hat{Y} = \{\hat{w}_1, \dots, \hat{w}_k\}$ with sequence of words $\hat{w}_i$ to concisely describe the crisis event and its evolution given the output from the timeline extraction task Textextracted." (p.7)
Q98	"We evaluate all timeline summarization models with ROUGE (Lin and Och, 2004)." (p.8)
Q153	"For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
WEB-2	Ecuador adult literacy ~94% (2022, World Bank) (https://www.theglobaleconomy.com/Ecuador/literacy_rate/)
WEB-3	Ecuador adult literacy ~96.3% (2022) (https://www.worldviewdata.com/countries/ecuador/literacy-rate)
WEB-5	https://iwgia.org/en/ecuador/5087-iw-2023-ecuador.html

Information Gaps

- Whether Spanish-tokenization-aware ROUGE or BERTScore variants have been validated for humanitarian summary evaluation; what specific genre-convention checks UN OCHA / Cruz Roja apply to situation-report summaries.

Requires Expert Verification

- Stylistic conventions and quality criteria for OCHA situation reports and Cruz Roja emergency appeals in Spanish that would need to be encoded in an adapted human evaluation rubric.

Remediation

Gap 1: ROUGE and human rubric are English-anchored and do not match UN OCHA situation report or Cruz Roja appeal genre conventions.

Recommendation: Adopt Spanish-tokenization-aware ROUGE plus a multilingual semantic similarity metric (e.g., BERTScore-multilingual); add genre-convention checks (numbered bullets, standardized headers, parroquia-level damnificado figures) to the human rubric; evaluate language-fidelity (does Spanish input produce Spanish output) as an explicit axis.